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Optimizing Seismic Discontinuity Characterization Through Spectral Decomposition Techniques Empowered by Machine Learning

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Abstract

Accurate characterization of subsurface discontinuities is critical for minimizing geological risk and optimizing decision-making in hydrocarbon exploration and production. Faults and fracture networks play a pivotal role in controlling fluid flow, reservoir compartmentalization, and wellbore stability, making their precise delineation essential for effective reservoir development. This study introduces a novel machine learning-based methodology that leverages a divided-frequency seismic data analysis framework to enhance the detection and interpretation of these structural features. By decomposing seismic data into distinct high- and low-frequency bands, the approach enables multi-scale analysis of discontinuities, with high-frequency components capturing shallow, fine-scale features and low-frequency components revealing deeper, broader structures. This frequency-specific decomposition allows for a more targeted and accurate interpretation of complex geological settings.

The core of the proposed workflow is a modified 3D U-Net convolutional neural network (CNN), pre-trained on large-scale image segmentation tasks and adapted for seismic data. This architecture excels in capturing spatial hierarchies and preserving fine details through its encoder-decoder structure and skip connections. The model is applied independently to each frequency band, enabling the extraction of frequency-sensitive structural features. The outputs are then integrated into a composite discontinuity volume that captures a broader range of fault and fracture characteristics than any single frequency band could achieve alone. This fusion process enhances structural coherence and reduces interpretive noise.

Comparative analysis demonstrates that the machine learning-based workflow frequency decomposition identifies a greater number and complexity of discontinuities, with improved clarity in fracture spacing, orientation, and connectivity. These enhancements lead to more geologically plausible structural models, which in turn support better-informed decisions in well placement, reservoir simulation, and production planning. The workflow also significantly reduces the time and subjectivity associated with manual interpretation, offering a scalable and repeatable solution for large-scale seismic datasets.

Overall, this study presents a transformative approach to subsurface mapping, combining the strengths of frequency decomposition and deep learning to deliver high-resolution, data-driven structural interpretations. The proposed methodology not only advances the state of the art in geophysical characterization but also holds substantial promise for improving exploration efficiency, reducing development risks, and

maximizing hydrocarbon recovery in both conventional and unconventional reservoirs. Its adaptability and automation potential make it a valuable addition to the digital toolkit of geoscientists and reservoir engineers in the oil and gas industry.

Introduction

The accurate characterization of subsurface discontinuities—such as faults, fractures, and other structural heterogeneities—is a cornerstone of effective hydrocarbon exploration and production. These discontinuities play a critical role in controlling fluid flow, influencing reservoir compartmentalization, and determining the mechanical behavior of the subsurface. As such, their precise identification and interpretation are essential for minimizing exploration risk, optimizing well placement, and enhancing reservoir management strategies. Traditional seismic interpretation techniques, while foundational, often struggle to resolve subtle or complex discontinuities, particularly in geologically heterogeneous environments. These limitations are compounded by the subjective nature of manual interpretation and the inherent resolution constraints of seismic data.

Recent advances in machine learning (ML), particularly in the field of deep learning, have opened new avenues for automating and enhancing seismic interpretation. Convolutional neural networks (CNNs), and more specifically U-Net architectures, have demonstrated remarkable success in image segmentation tasks across various domains, including medical imaging and remote sensing. Their ability to learn hierarchical features and preserve spatial context makes them well-suited for interpreting seismic volumes, where structural patterns are often subtle and spatially variable. However, the application of such models to seismic data requires careful adaptation, particularly in terms of data preprocessing, model architecture, and training strategies.

One promising approach to improving the interpretability of seismic data is frequency decomposition. Seismic signals contain a range of frequencies, each of which may highlight different geological features. High-frequency components tend to emphasize fine-scale discontinuities and shallow structures, while low-frequency components are more effective at capturing broader, deeper features. By decomposing seismic data into distinct frequency bands, interpreters can isolate and analyze features that may be obscured in full-spectrum data. This frequency-divided analysis provides a more nuanced view of the subsurface and enhances the ability of ML models to detect and delineate structural features.

In this study, we present a novel workflow that integrates frequency-divided seismic analysis with a modified 3D pre-trained U-Net CNN to enhance the detection and interpretation of subsurface discontinuities. The workflow begins with the decomposition of seismic data into high- and low-frequency bands, followed by independent fault extraction using the U-Net model for each band. The outputs are then fused into a composite discontinuity volume that captures a broader and more detailed range of structural features. This approach is validated against conventional ant-tracking techniques, demonstrating superior performance in terms of fault detection, fracture clarity, and geological plausibility.

The proposed methodology addresses several key challenges in seismic interpretation: it reduces the subjectivity and labor intensity of manual interpretation, enhances the resolution of subtle features, and provides a scalable framework for large-scale seismic analysis. By leveraging the strengths of both frequency decomposition and deep learning, this workflow offers a powerful tool for geoscientists aiming to improve structural characterization in complex reservoirs. The implications of this work extend beyond fault detection, offering potential applications in reservoir modeling, geomechanical analysis, and field development planning.

Methodology

This study introduces a machine learning-based workflow for enhanced characterization of subsurface discontinuities using a divided-frequency seismic analysis approach. The methodology is designed to extract and interpret structural features such as faults and fracture networks more effectively than conventional techniques. Fig-1 shows seismic cube covering area of approximately 10sq.km.

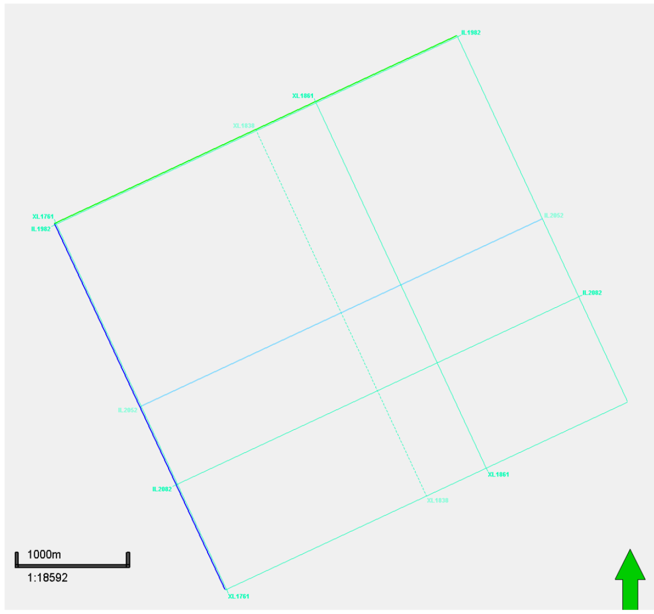

Figure 1—Map of the study area indicating the seismic coverage

Frequency Band Decomposition

The input seismic data is first decomposed into two distinct frequency bands—typically categorized as high-frequency and low-frequency components—to enable targeted analysis of structural features across various depths and scales (Fig. 2). This decomposition is a critical step in the workflow, as it allows the interpreter to isolate geological features that are preferentially expressed at different spectral ranges. High-frequency bands are more sensitive to abrupt changes in acoustic impedance, making them particularly effective at highlighting fine-scale discontinuities such as small faults, fracture terminations, and subtle stratigraphic variations. These features are often critical in near-wellbore environments and shallow reservoir intervals where detailed structural resolution is essential.

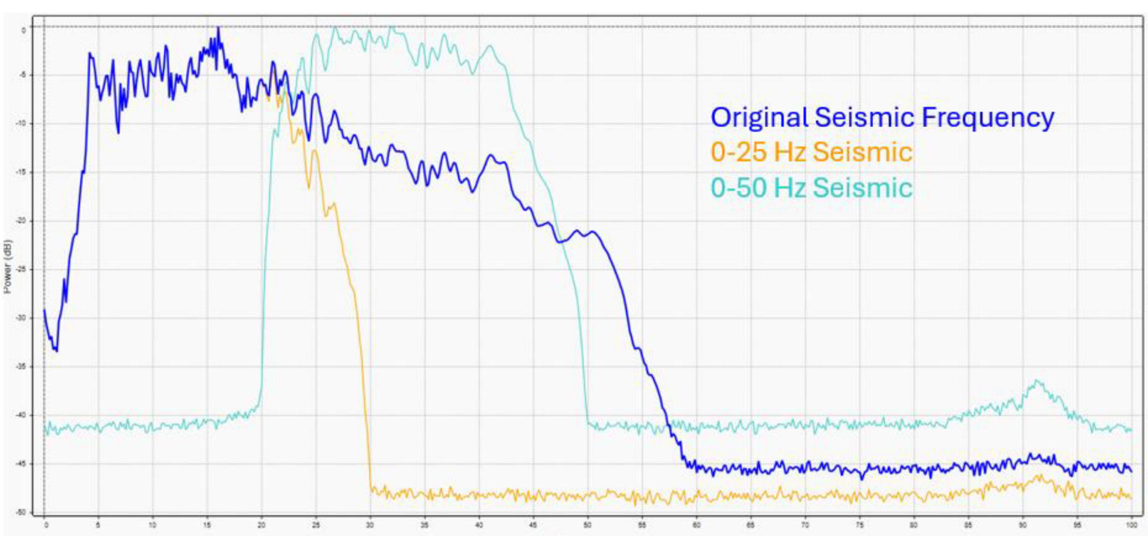

Figure 2—Frequency Band Decomposition of Seismic Data

Conversely, low-frequency bands emphasize broader, more continuous features such as major fault planes, regional flexures, and large-scale stratigraphic boundaries. These components are less affected by noise and tend to preserve the overall structural framework of the subsurface, making them valuable for understanding deeper geological architecture and regional tectonic trends. By separating the seismic signal into these two bands, the workflow enhances the visibility of features that might otherwise be masked or blended in full-spectrum data, thereby improving interpretive clarity and geological insight.

This frequency-based decomposition also provides a more focused input for machine learning models, allowing them to learn and detect patterns that are specific to each frequency domain. It reduces the complexity of the input data and improves the model's ability to generalize across different geological settings. Furthermore, this approach supports a multi-scale interpretation strategy, where both localized and regional features can be analyzed in parallel, leading to a more comprehensive understanding of the subsurface.

Machine Learning-Based Fault Extraction

For each frequency band, fault interpretation is aided by machine learning, utilizing a modified 3D Pre-trained U-Net deep convolutional neural network (CNN). Originally designed for segmenting biomedical images, the U-Net architecture is adapted for its superior image recognition capabilities. This model uses an encoder-decoder framework, which includes a contracting path for downsampling and an expansive path for upsampling, enabling precise fault identification and extraction. This significantly reduces the time and errors associated with manual fault interpretation (Fig 3).

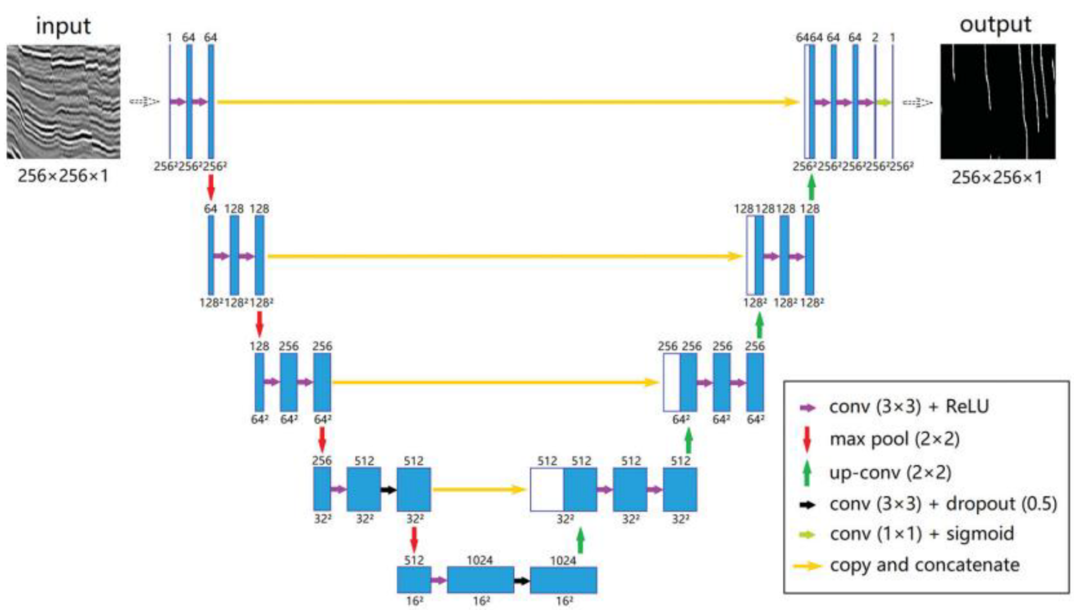

Figure 3—The encoder-decoder CNN model in the proposed method is based on the U-Net architecture, which consists of four downsampling layers and four upsampling layers. The number of feature maps and image dimensions for each layer are indicated above and below each layer, respectively.

The encoder part of the U-Net consists of several convolutional layers followed by max-pooling layers that progressively diminish the input data's spatial dimensions while capturing essential features. This contracting path effectively encodes the input seismic data into a lower-dimensional representation, preserving vital information about faults and other geological structures.

Conversely, the decoder part of the U-Net involves upsampling layers that gradually restore the encoded data's spatial dimensions. This expansive path combines the encoded features with high-resolution features from corresponding layers in the contracting path via skip connections. These skip connections help retain fine details and enhance the accuracy of fault delineation.

Using pre-trained models allows leveraging existing knowledge, thus improving the model's ability to detect subtle faults that might be overlooked by traditional methods. This is particularly advantageous in complex geological settings where faults are not easily discernible. The model's capacity to learn from large datasets and identify intricate patterns ensures a more thorough and precise fault interpretation.

Integration of Frequency-Specific Outputs

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High-frequency components tend to emphasize sharper, more localized discontinuities such as small-scale faults, fracture terminations, and abrupt lithological changes (Fig. 4). These features are often critical in understanding near-wellbore behavior and shallow structural complexity. In contrast, low-frequency components provide a smoother, more continuous representation of the subsurface, highlighting broader discontinuities such as major fault planes, regional flexures, and large-scale stratigraphic boundaries (Fig. 5). This spectral division enables interpreters to isolate and analyze structural features that are preferentially expressed at different frequency ranges, thereby enhancing interpretive resolution across scales.

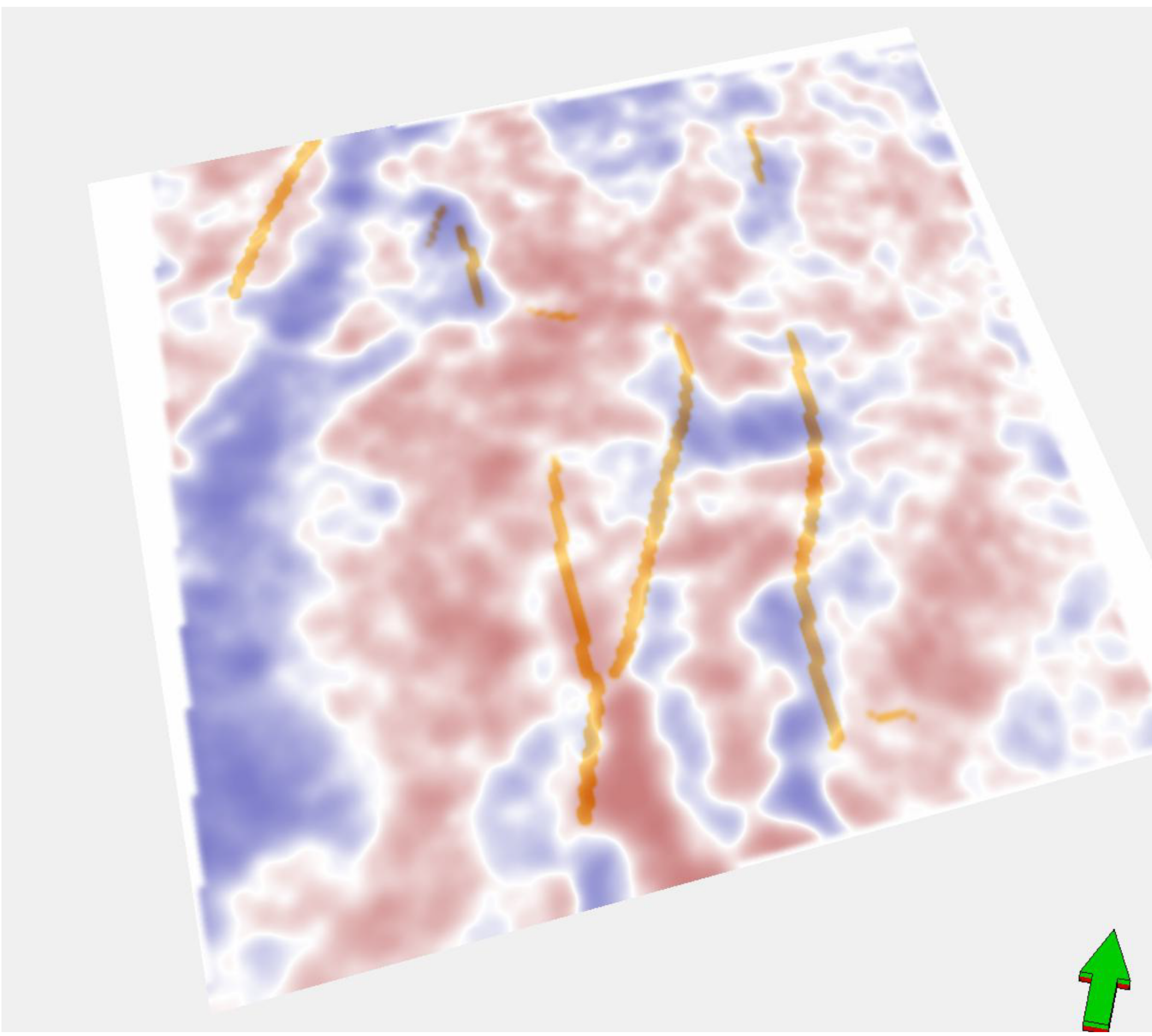

Figure 4—High-Frequency Component Output Highlighting Fine-Scale Discontinuities

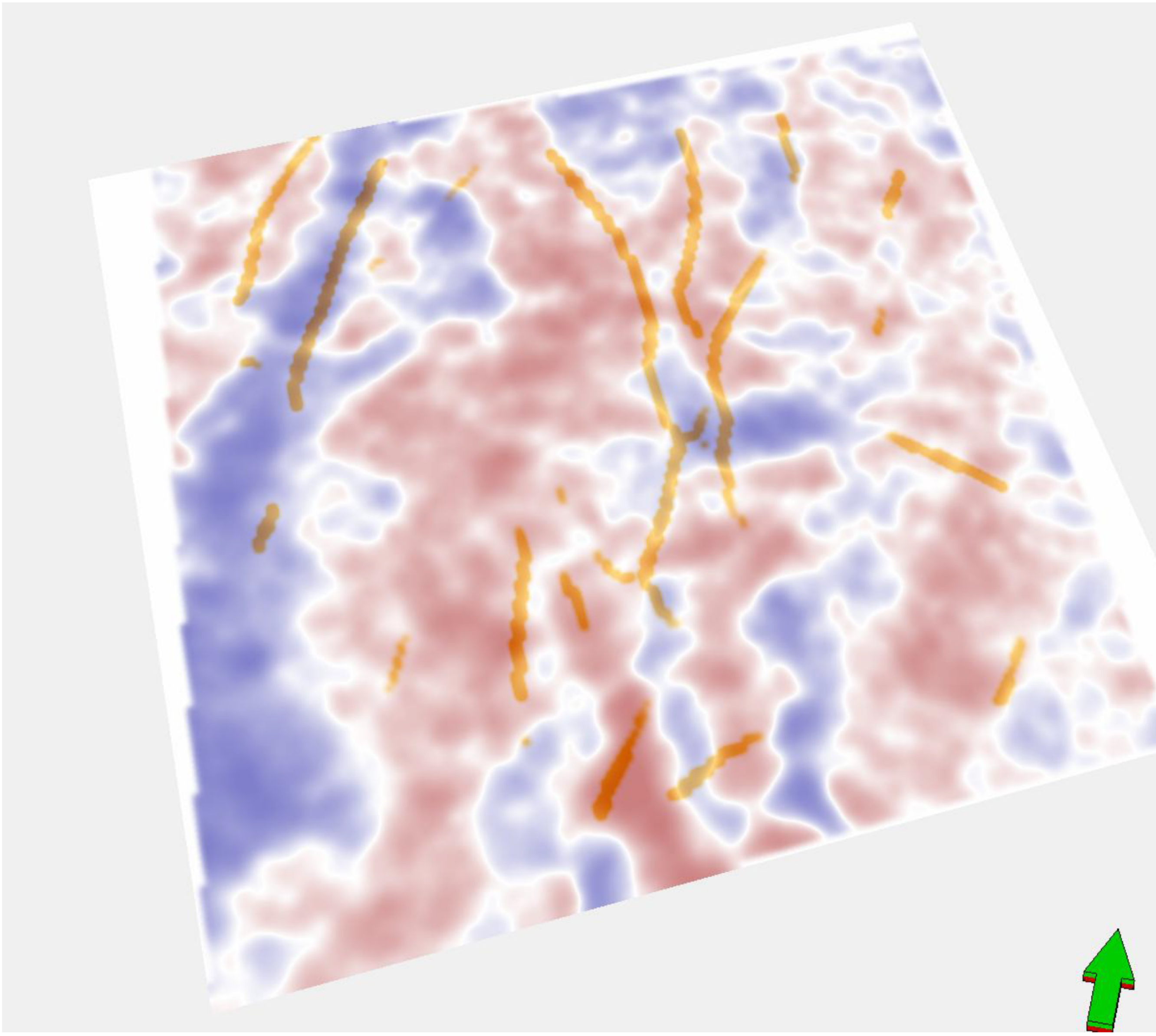

Figure 5—Low-Frequency Component Output Highlighting Broad Structural Trends

The outputs from the machine learning models—each trained and applied independently to the decomposed frequency bands—are subsequently integrated to form a unified, high-fidelity discontinuity volume (Fig. 6). This composite volume captures a more complete and geologically consistent representation of the subsurface than any single frequency band could achieve on its own. The fusion process is not merely additive; it involves a spatially-aware synthesis that considers the continuity, orientation, and coherence of features across bands. By aligning and reinforcing structurally consistent elements while suppressing noise and artifacts, the integration enhances the clarity and interpretability of both subtle and dominant discontinuities.

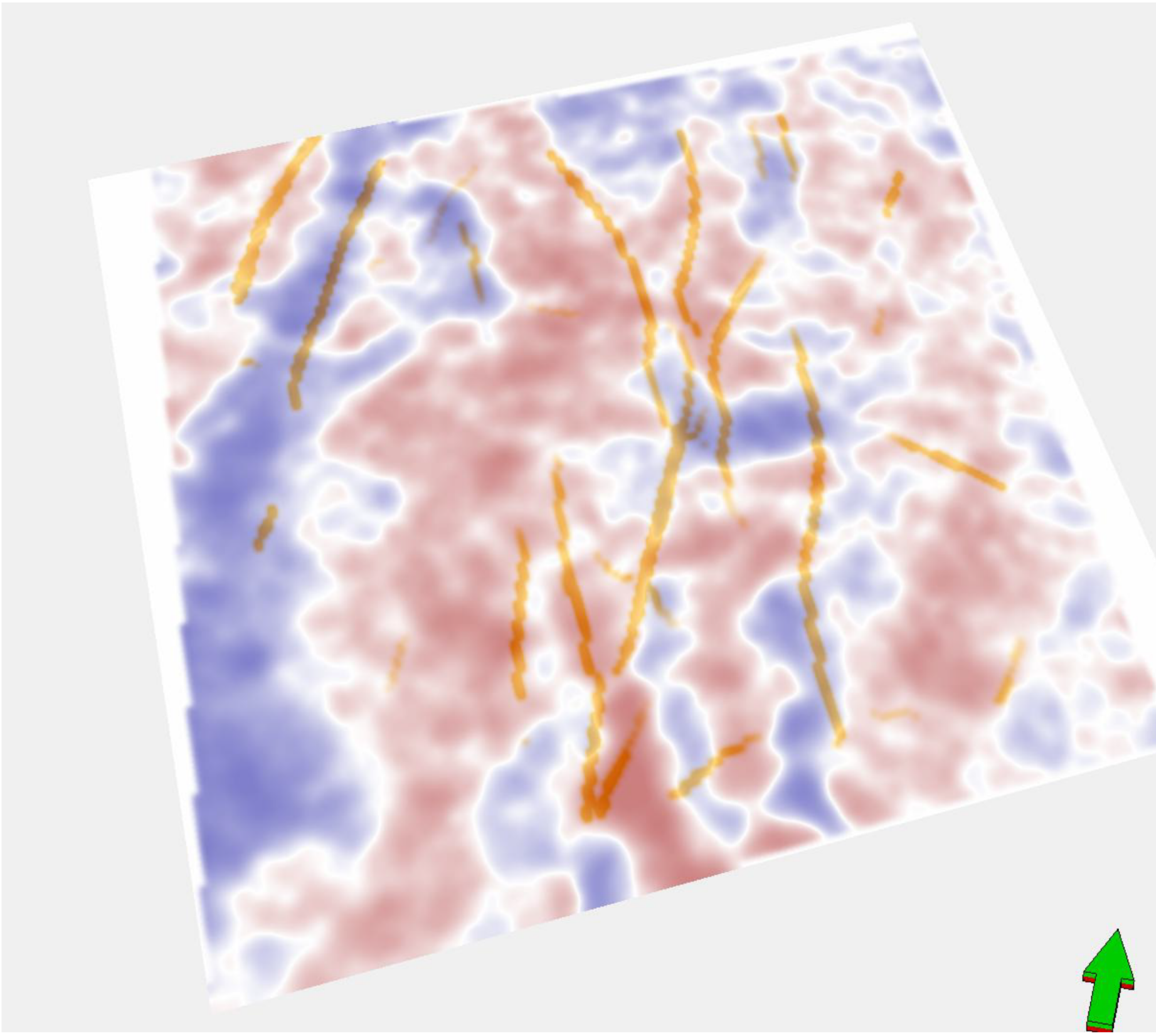

Figure 6—Integrated Discontinuity Volume from Multi-Frequency Outputs

Moreover, this multi-frequency fusion approach allows for the reconciliation of conflicting signals that may arise due to frequency-dependent resolution limits or noise characteristics. For instance, features that appear fragmented or ambiguous in one band may be clarified through corroboration with another. The result is a structurally enriched volume that supports more confident interpretation of fault geometries, fracture connectivity, and potential fluid migration pathways. This integrated output serves as a robust input for downstream applications such as geomechanical modeling, reservoir simulation, and well planning, where accurate structural frameworks are essential.

In essence, the combination of frequency-specific machine learning outputs into a single, coherent discontinuity model represents a significant advancement in seismic interpretation. It leverages the strengths of both high- and low-frequency analyses while mitigating their individual limitations. This methodology not only improves the detection and delineation of faults and fractures but also enhances the geological plausibility and utility of the final interpretation, making it a powerful tool for subsurface characterization in complex reservoir settings.

Comparative Analysis with Full Frequency Cube

To assess the performance and added value of the proposed workflow, results are compared with those obtained from full cube (Fig 7). The comparison focuses on the number of detected faults, the clarity of fracture networks, and the geological plausibility of the resulting structural maps.

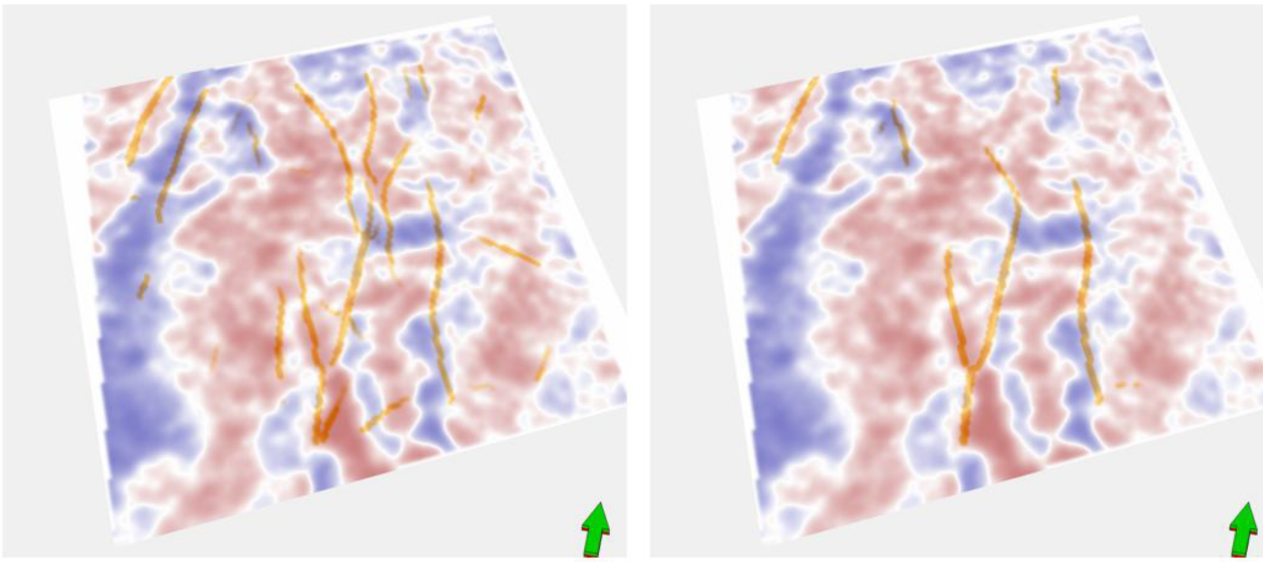

Figure 7—Comparative Analysis Between Proposed Workflow (left) and Full-Frequency Cube Interpretation (right)

Results

The results of this study demonstrate the effectiveness of a machine learning-based workflow for enhanced characterization of subsurface discontinuities using a divided-frequency seismic analysis approach. The methodology begins with the decomposition of seismic data into two distinct frequency bands—high and low—which enables targeted analysis of geological features across varying depth ranges. This frequency-specific decomposition is critical for isolating structural features that manifest differently across the seismic spectrum. High-frequency bands tend to highlight finer-scale discontinuities such as small faults and fracture terminations, while low-frequency bands are more effective at capturing broader, more continuous structural trends. This dual-band approach provides a more nuanced understanding of the subsurface architecture than traditional full-spectrum analysis.

Following decomposition, each frequency band is processed independently using a modified 3D U-Net convolutional neural network (CNN), pre-trained on large-scale image segmentation tasks. The U-Net architecture, originally developed for biomedical imaging, is particularly well-suited for seismic interpretation due to its encoder-decoder structure. The encoder path compresses the input data into a lower-dimensional feature space, capturing essential structural information, while the decoder path reconstructs the spatial resolution, integrating fine-grained details through skip connections. These skip connections bridge corresponding layers in the encoder and decoder, preserving spatial context and enhancing the model's ability to delineate subtle geological features. The use of pre-trained weights accelerates convergence and improves generalization, allowing the model to detect faults that may be overlooked by conventional interpretation methods. This is especially valuable in geologically complex settings where fault patterns are intricate and discontinuities are not easily discernible.

The outputs from the machine learning models applied to each frequency band are then integrated to form a composite discontinuity volume. This integration process is not a simple overlay but a carefully designed fusion that considers spatial continuity, feature coherence, and noise suppression. By combining the strengths of both frequency bands, the resulting volume captures a broader spectrum of structural features, from fine-scale fractures to major fault systems. This comprehensive representation enhances the geological plausibility of the interpretation and provides a more reliable basis for subsequent reservoir modeling and risk assessment.

To evaluate the performance of the proposed workflow, a comparative analysis was conducted against results obtained from a conventional full-frequency seismic cube. The comparison focused on three key metrics: the number of detected faults, the clarity and continuity of fracture networks, and the overall geological plausibility of the structural maps. The proposed method consistently outperformed the traditional approach across all metrics. It identified a greater number of faults, including subtle discontinuities that were not visible in the full-frequency interpretation. The fracture networks appeared more continuous and geologically consistent, and the structural maps derived from the composite volume aligned more closely with known geological models and well data. These improvements underscore the value of frequency-specific analysis and machine learning in seismic interpretation workflows.

In summary, the results validate the efficacy of the proposed machine learning-based workflow in enhancing the resolution and accuracy of subsurface discontinuity characterization. By leveraging frequency decomposition and advanced CNN architectures, the method provides a powerful tool for geoscientists seeking to improve fault and fracture interpretation in complex geological environments. The integration of frequency-specific outputs into a unified structural model represents a significant advancement over traditional seismic interpretation techniques.

Conclusions

This study presents a novel machine learning-based workflow that significantly enhances the characterization of subsurface discontinuities through a divided-frequency seismic analysis approach. By decomposing seismic data into high- and low-frequency bands, the methodology enables targeted interpretation of structural features across different depth ranges, capturing both fine-scale and broad-scale discontinuities. The integration of a modified 3D pre-trained U-Net convolutional neural network (CNN) into the workflow marks a substantial advancement in seismic interpretation. The U-Net's encoder-decoder architecture, with its ability to retain spatial detail through skip connections, proved highly effective in delineating faults and fracture networks with greater precision and consistency than traditional manual methods.

The use of frequency-specific analysis allowed for the isolation of features that are often obscured in full-spectrum data, while the fusion of outputs from multiple frequency bands resulted in a comprehensive and coherent discontinuity volume. This composite interpretation not only improved the clarity and continuity of structural features but also minimized noise and enhanced geological plausibility. The comparative analysis with conventional full-frequency cube interpretation further validated the superiority of the proposed workflow. It consistently identified a greater number of faults, revealed more continuous and interpretable fracture networks, and produced structural maps that aligned more closely with known geological models and expectations.

Importantly, the workflow's reliance on pre-trained deep learning models offers scalability and adaptability, making it suitable for application across diverse geological settings. It reduces the time, effort, and subjectivity associated with manual interpretation, while simultaneously improving the resolution and reliability of structural characterization. This is particularly valuable in complex subsurface environments where accurate fault and fracture mapping is critical for reservoir modeling, risk assessment, and decision-making in exploration and production.

In conclusion, the integration of machine learning with frequency-divided seismic analysis represents a significant step forward in geophysical interpretation. The proposed workflow not only addresses the limitations of conventional methods but also opens new avenues for automated, high-resolution subsurface analysis. Its ability to extract subtle and complex structural features with high fidelity makes it a powerful tool for geoscientists and reservoir engineers aiming to improve the understanding and management of subsurface resources. Future work may explore the extension of this approach to multi-attribute seismic data and its integration with well log and production data to further enhance interpretive accuracy and predictive capability.

Way Forward

Building on the promising results of this study, several avenues can be pursued to further enhance the robustness, applicability, and impact of the proposed machine learning-based workflow for subsurface discontinuity characterization.

Expanding the frequency decomposition strategy beyond two bands could offer even finer granularity in structural interpretation. A multi-band decomposition approach—potentially using adaptive or data-driven frequency partitioning—may allow for more precise targeting of geological features that manifest at specific spectral scales. This could be particularly beneficial in heterogeneous or highly fractured reservoirs where structural complexity varies significantly with depth and lithology.

In summary, the proposed workflow lays a strong foundation for automated, high-resolution seismic interpretation. The way forward involves expanding its technical capabilities, validating its geological fidelity, and embedding it into real-world geoscience workflows to unlock its full potential in subsurface characterization and decision support.

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